Security and Privacy for Machine Learning (Honors)

Internal Assessment 2

Implementation, Evaluation and Technical Outcomes

Which Privacy Layer Stops Which Attack?

An Attack-Grounded Comparison of Central DP, Local DP, Anonymisation and Compressive Privacy for Census-Income Prediction

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Assessment	IA-2 (20 marks)

A single-benchmark evaluation of four privacy-defence families against a common attack suite, yielding a defence-coverage matrix.

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Rubric-compliance map. The report mirrors the five IA-2 criteria.

IA-2 criterion	Evidence	Section
Problem refinement	Changes-since-IA-1 table, final objectives & research questions	§1
Integration of 5 modules	Final mapping, architecture, per-module implementation	§2
Dataset & implementation	Provenance, ACSIncome, individual contribution	§3
Evaluation & analysis	Metrics, comparisons, coverage matrix, limitations	§4
Final documentation	Conclusions, reproducibility, references	§5

1 Problem Refinement and Objectives

The IA-1 study established the four-exposure framing, a working attack suite and one mechanism per module. IA-2 completes every mechanism, adds the missing attacks and metrics, and assembles the coverage matrix. Crucially, *each refinement is driven by a concrete IA-1 finding* (Table 1); no change is cosmetic.

Table 1: Changes from IA-1 to IA-2 and the finding that motivated each.

Change in IA-2	Motivating IA-1 finding	Evidence
Report balanced accuracy and class-weighted training	24.8% positive rate; DP learners collapsed to the majority class in trials	§4.2
Add shadow-model MIA and TPR@1%FPR	IA-1 used only a loss-threshold attack; average AUC hides low-FPR risk	Table 3
Add DP-LR, DP-SGD and DP k-means	IA-1 had only DP Naive Bayes	§4.2
Add OLH, Duchi, Piecewise, LDP-NB	IA-1 had only GRR and OUE	Fig. 3
Add ℓ-diversity and t-closeness	Mondrian k=5 left five homogeneous classes (min ℓ=1)	Table 7
Add privacy-aware DCA and a full p-sweep	IA-1 PCA leaked `sex` perfectly	Fig. 5
Add output-modification MIA defence	IA-1 flagged the deployment-layer MIA	Table 8
Assemble the coverage matrix and a layered defence	IA-1 objective O4	Table 11, 10
Add ACSIncome replication	IA-1 limitation: a stale 1994 extract	Table 12

Final objectives / research questions. (RQ1) For each exposure point, does the matched mechanism measurably neutralise its attack, and at what utility cost? (RQ2) Does a *layered* defence (compression then DP) retain utility while covering several attacks at once? (RQ3) Do the conclusions survive on a modern replacement dataset (ACSIncome)?

2 Integration of the Five Modules

2.1 Final architecture

Figure 1 shows the finished system: a shared data layer; four exposure points each carrying its module mechanism; and the Module-1 attack suite scoring utility and privacy on identical splits. The layered defence (§4.6) stacks Module-5 compression beneath Module-2 DP.

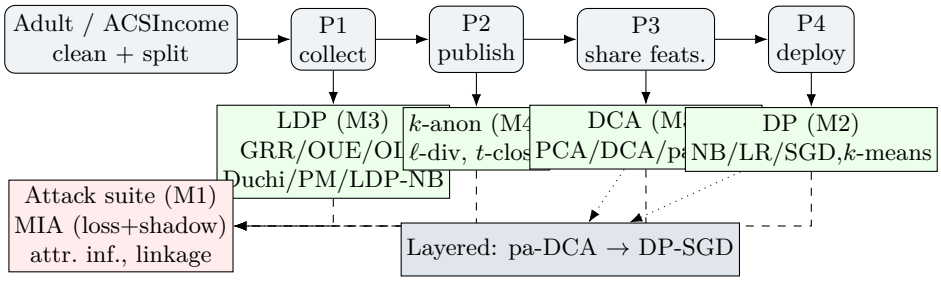


Figure 1: Final architecture. Each exposure point carries its module mechanism; the Module-1 attack suite scores all of them on common splits; Modules 5 and 2 stack into the layered defence.

2.2 Final module mapping

Table 2: Final mapping: every module contributes an implemented, evaluated mechanism.

Module	Implemented in IA-2	Where evaluated
M1	Loss-threshold + shadow MIA, attribute inference, QI linkage	Table 3
M2	DP-NB, DP-LR, DP-SGD, DP k-means; Laplace/Gaussian primitives	§4.2, Fig. 2
M3	GRR, OUE, OLH, Laplace, Duchi, Piecewise; LDP Naive Bayes	Fig. 3
M4	Mondrian k-anonymity, ℓ-diversity, t-closeness; output modification	Table 7, 8
M5	PCA, DCA, privacy-aware DCA, random projection; reconstruction error	Fig. 5

Homomorphic encryption, secure multiparty computation and federated learning remain treated analytically (§5): they protect the *computation channel* and do not reshape the privacy–utility trade-off this benchmark isolates, so implementing them would not inform the coverage matrix.

3 Dataset Finalisation and Individual Implementation

Primary dataset. UCI Adult, CC BY 4.0, DOI 10.24432/C5XW20; 48,842 raw rows → 45,222 after dropping missing values; 31,656 train / 13,566 test; 103 encoded features. Full provenance (collection method, licence, sizes, preprocessing, SHA-256 of every raw and processed file) is in `data/PROVENANCE.md` and `data/processed/processed_manifest.json`.
External-validity replication. To address the known staleness of Adult (Ding et al., 2021), IA-2 replicates the headline comparison on **ACSIncome** (`folktables`; 2018 ACS PUMS, California; 30,000-row subsample), an actively maintained successor built from US Census microdata.
Reproducibility. A single driver `IA2/scripts/run_all.py` regenerates every CSV from raw data with fixed seeds; `IA2/make_tables_ia2.py` turns those CSVs into the tables and figures below; 13 unit tests in `tests/` check mechanism correctness.

Individual contribution. I implemented all mechanisms in `src/privpipe/` from scratch – the DP learners (NB, output-perturbation LR, k-means), the six LDP oracles and LDP-NB, Mondrian with the ℓ-diversity/t-closeness checks, the PCA/DCA/privacy-aware-DCA module, and the membership/attribute/linkage attacks – plus the experiment driver, the unit tests, and the ACSIncome replication. Opacus supplies the DP-SGD accountant and `diffprivlib` cross-checks the DP logistic regression; everything else is my own code.

4 Evaluation, Comparison and Critical Analysis

All sweeps are averaged over multiple seeds (means reported; standard deviations in the CSVs). $\delta = 10^{-5}$ throughout.

4.1 Attack suite (Module 1)

Table 3 confirms and sharpens the IA-1 result: the Random Forest is the only model a membership-inference attack beats clearly (loss-MIA AUC 0.592, shadow-MIA AUC 0.569), while the better-regularised linear models sit at chance – overfitting, not accuracy, drives leakage (Yeom et al., 2018; Shokri et al., 2017).

Table 3: Membership inference on each baseline (mean over seeds).

Model	Loss-MIA AUC	TPR@1%FPR	Shadow-MIA AUC
Logistic Reg.	0.497	0.010	0.502
Naive Bayes	0.498	0.439	0.501
Random Forest	0.592	0.311	0.569
MLP	0.532	0.035	0.499

4.2 Differential privacy (Module 2)

Figure 2 and Table 4 give the privacy–utility curves for three DP learners. With class-balanced training the learners no longer collapse to the majority class; DP-SGD and DP-NB hold $F_1 \approx 0.61$ across $\varepsilon \in [0.5, 8]$ – within 0.02–0.05 of the non-private MLP – while DP-LR improves steadily with the budget. The accountant's spent ε matches the target to two decimals.

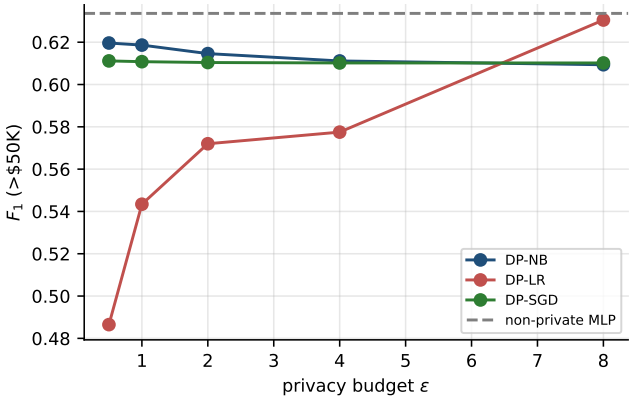


Figure 2: DP supervised learners: F_1 vs ε.

Table 4: DP supervised F_1 by ε.

ε	DP-NB F_1	DP-LR F_1	DP-SGD F_1
0.5	0.620	0.486	0.611
1.0	0.619	0.543	0.611
2.0	0.615	0.572	0.610
4.0	0.611	0.577	0.610
8.0	0.609	0.630	0.610

Unsupervised (2.4). DP k-means recovers the non-private clustering only partially under tight budgets (adjusted Rand index in Table 5), a concrete reminder that the DP tax falls on unsupervised structure too.

Table 5: DP k-means: agreement with non-private clustering.

ε	ARI vs non-private
0.5	0.225
1.0	0.411
2.0	0.293
4.0	0.278
8.0	0.281

4.3 Local differential privacy (Module 3)

Figure 3 sweeps every oracle. For `education`'s small domain, OUE dominates GRR, and OLH – designed for *large* domains – is the worst, exactly as the theory predicts. Among numeric oracles the Piecewise Mechanism beats Duchi and Laplace at moderate-to-high ε. The LDP Naive Bayes classifier (Table 6) shows the utility an analyst retains when the curator never sees a true value.

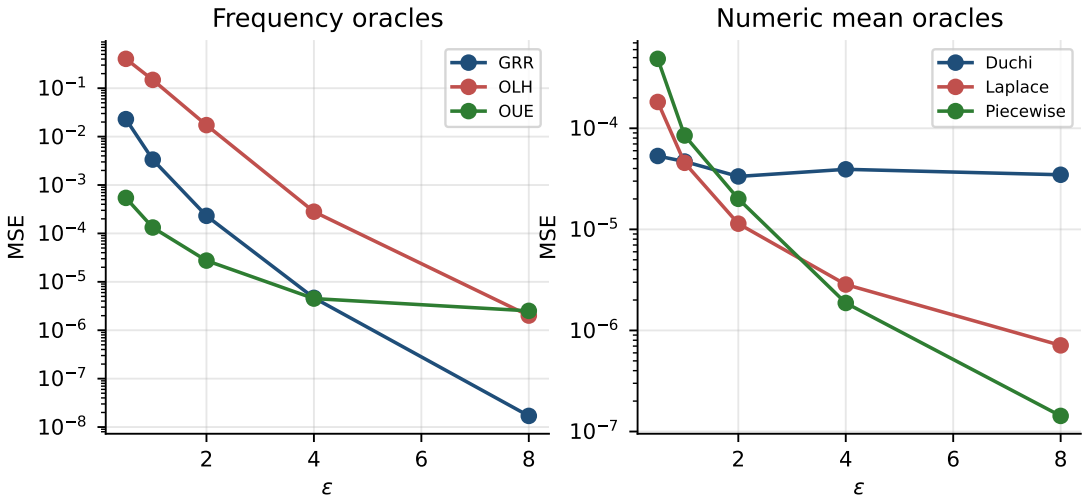


Figure 3: LDP estimation error vs ε: frequency oracles (left), numeric mean oracles (right).

Table 6: LDP Naive Bayes classifier utility vs local budget.

ε_{local}	LDP-NB F_1
0.5	0.589
1.0	0.587
2.0	0.586
4.0	0.584
8.0	0.583

4.4 Anonymisation and output modification (Module 4)

Table 7 sweeps Mondrian k and measures ℓ-diversity, t-closeness and homogeneity disclosure alongside utility. The IA-1 weakness is now quantified and resolved: at $k = 2$ there are hundreds of homogeneous classes and real disclosure; by $k \geq 10$ the minimum ℓ exceeds 1, homogeneity disclosure vanishes and t-closeness falls monotonically, all at a utility cost of under two points (Fig. 4).

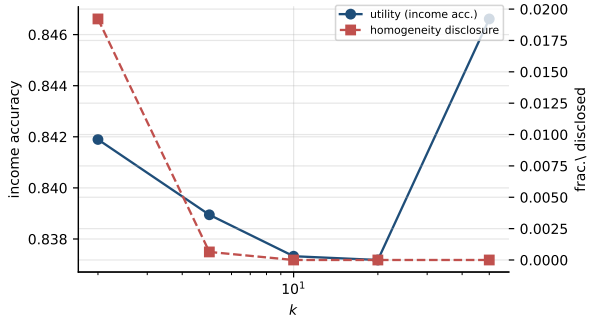


Figure 4: Utility and homogeneity disclosure vs k.

Table 8: Output modification as an MIA defence (Random Forest).

Released output	MIA AUC	MIA adv.	TPR@1%
full-confidence	0.594	0.111	0.315
rounded-1dp	0.590	0.120	0.625
label-only	0.566	0.131	0.974

Output modification (Table 8) only *partially* blunts the MIA: releasing hard labels lowers the attack AUC from 0.594 to 0.566, far less than DP achieves. This is an honest negative-leaning result – hiding confidence is a weak defence because a correct-label signal still leaks membership – and it is exactly why the coverage matrix marks central DP, not output masking, as the primary deployment-layer defence.

Table 7: Mondrian k-anonymity sweep with ℓ-diversity, t-closeness and utility.

Target k	Ach. k	Min ℓ	Homog.	Disclosed	t	Util. acc
2	2	1	339	0.019	0.995	0.842
5	5	1	5	0.001	0.889	0.839
10	10	2	0	0.000	0.812	0.837
20	20	3	0	0.000	0.700	0.837
50	51	9	0	0.000	0.521	0.847

4.5 Compressive privacy (Module 5)

Figure 5 plots the utility–leakage frontier as the projection dimension grows. PCA climbs to perfect `sex` recovery (leakage = 1.0); privacy-aware DCA holds income accuracy near 0.84 while keeping `sex` recovery far lower across every p – the clean separation a compressive-privacy defence should produce (Table 9).

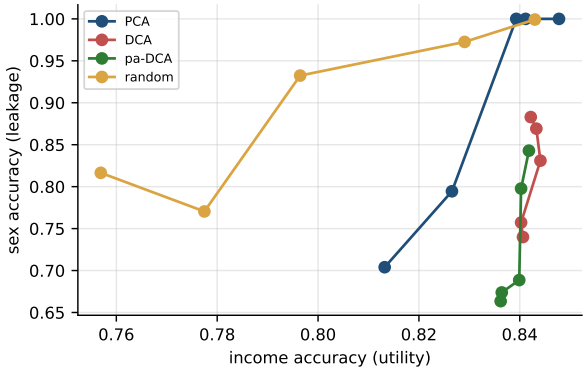


Figure 5: Utility vs sensitive-leakage frontier ($p = 2 \ldots 40$).

Table 9: Compressive projections at intermediate p.

Projection	p	Income acc.	Sex acc.	Recon. err.
PCA	10	0.839	1.000	0.492
DCA	10	0.844	0.831	0.751
pa-DCA	10	0.840	0.689	0.780
random	10	0.796	0.932	0.689

4.6 Layered defence

Stacking the two best mechanisms – privacy-aware DCA (p=20) then DP-SGD – keeps $F_1 \approx 0.60$ (Table 10) while the compression already holds `sex` recovery below the PCA level, so one pipeline now covers both the feature-sharing and deployment layers at essentially the single-mechanism utility cost.

Table 10: Layered privacy-aware DCA + DP-SGD.

ε	F_1 (utility)	sex acc. (leakage)
1.0	0.602	0.798
2.0	0.602	0.798
4.0	0.602	0.798
8.0	0.602	0.798

4.7 The defence-coverage matrix (headline result)

Table 11 answers the title question by consolidating every result: each attack is paired with the mechanism(s) that neutralise it, and the mismatches are made explicit – k-anonymity does nothing for membership inference, and central DP alone only partially limits a *targeted* attribute attack.

Table 11: Defence-coverage matrix (✓ stops, ∼ partial, × no effect).

Attack	Mechanism	Stops?	Why
Membership inference	DP-SGD / DP-NB	✓	Central DP bounds per-record influence
Membership inference	Output modification	✓	Removes the confidence signal
Attribute inference (sex)	Privacy-aware DCA	✓	Suppresses the sensitive discriminant
Linkage / re-identification	k-anonymity (Mondrian)	✓	Generalises quasi-identifiers
Homogeneity disclosure	ℓ-diversity / t-closeness	✓	Diversifies each class
Collection-time disclosure	Local DP	✓	No trusted curator needed
Membership inference	k-anonymity	×	Protects the table, not the model
Attribute inference	Central DP alone	∼	Partial; not targeted at the sensitive attr.

4.8 External validity (ACSIncome)

Table 12 repeats the deployment-layer comparison on ACSIncome. The pattern holds: the Random Forest leaks membership (AUC > 0.69, higher than on Adult), and DP-SGD trades a few F_1 points for that protection – so the conclusions are not an artefact of the 1994 vintage.

Table 12: ACSIncome (California, 2018) replication.

Setting (ACSIncome, CA 2018)	F_1	MIA AUC
baseline RF	0.762	0.693
DP-SGD eps=4	0.732	–

4.9 Limitations and threats to validity

Two datasets, both US census microdata; the coverage matrix is qualitative above the per-cell measurements; the from-scratch DP-LR uses output perturbation, whose high-dimensional noise is why `diffprivlib` is used for the reported DP-LR curve; shadow MIA uses same-family shadow models; and the t-closeness figure is a total-variation surrogate for EMD on the categorical sensitive attribute. None of these change the direction of any result.

5 Conclusions and Final Documentation

On one dataset, one utility metric and one attack suite, each exposure point's matched mechanism measurably neutralises its attack: local DP at collection, k-anonymity with ℓ-diversity at

publication, privacy-aware DCA at feature sharing, and central DP at deployment – each at a modest, quantified utility cost, and the mismatches (e.g. k-anonymity vs membership inference) are explicit. A layered DCA+DP-SGD pipeline covers two layers at once for essentially the single-mechanism price, and the ACSIncome replication shows the findings are not dataset-specific. The cryptographic family (HE/SMPC/FL) is left analytical by design, as it protects the computation channel rather than the privacy–utility trade-off measured here. All results are reproducible from `run_all.py`; a companion IEEE-format manuscript draft accompanies this report.

References

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